

MOHAN KUMAR KUMAR

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Research Scientist
Meta Reality Labs

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Education

Georgia Institute of Technology	Ph.D. in Computer Science Specialization: Systems Thesis: Taming Latency In Data Center Applications; Advisor: Dr. Taesoo Kim	2014–2019, Atlanta, GA
Georgia Institute of Technology	M.S. in Computer Science Specialization: Systems	2014–2019, Atlanta, GA
University of Madras	B.E. in Computer Science	Chennai, India

Current Role

At Meta Reality Labs, I serve as a Research Scientist focused on accelerating machine learning inference on embedded devices. My work encompasses optimizing and deploying diverse ML architectures—including CNNs, RCNNs, RNNs, and transformers—onto resource-constrained hardware platforms. I leverage PyTorch and related frameworks to develop efficient model implementations and quantization techniques for edge deployment.

I also partner with SoC vendors to architect NPU/eNPU specifications that enable the development of more efficient and effective hardware solutions tailored for on-device ML inference applications.

Peer-Reviewed Publications

1. **ECOTLB: Eventually Consistent TLBs**
Steffen Maass*, **Mohan Kumar***, Taesoo Kim, Tushar Krishna, and Abhishek Bhattacharjee.
ACM Transactions on Architecture and Code Optimization (TACO), 2020.
2. **SOLROS: A Data-Centric Operating System Architecture for Heterogeneous Computing**
Changwoo Min, Woon-Hak Kang, **Mohan Kumar**, Sanidhya Kashyap, Steffen Maass, Heeseung Jo, and Taesoo Kim.
Proceedings of EuroSys 2018, Porto, Portugal, April 2018.
3. **LATR: Lazy Translation Coherence**
Mohan Kumar*, Steffen Maass*, Sanidhya Kashyap, Jan Vesely, Zi Yan, Taesoo Kim, Abhishek Bhattacharjee, and Tushar Krishna.
Proceedings of ASPLOS 2018, Williamsburg, VA, USA, March 2018.
4. **Mosaic: Processing a Trillion-Edge Graph on a Single Machine**
Steffen Maass, Changwoo Min, Sanidhya Kashyap, Woonhak Kang, **Mohan Kumar**, and Taesoo Kim.
Proceedings of EuroSys 2017, Belgrade, Serbia, April 2017. **[Best Student Paper]**
5. **TCP Ordo: The Cost of Ordered Processing in TCP Servers**
Mohan Kumar and Ada Gavrilovska.
Proceedings of INFOCOM 2016, San Francisco, CA, April 2016.
6. **S-NFV: Securing NFV States by Using SGX**
Ming-Wei Shih, **Mohan Kumar**, Taesoo Kim, and Ada Gavrilovska.
Proceedings of SDN-NFV Security 2016, New Orleans, LA, March 2016. **[Best Paper]**

Posters and Workshops

1. | **mKPAC: Kernel Packet Processing for Manycore Systems**
Ramneek, **Mohan Kumar**, Taesoo Kim, and Sungin Jung.
Middleware 2018 Poster, Rennes, France, December 2018.
2. | **Network Function Fault Isolation in a Single Address Space**
Mohan Kumar, Steffen Maass, and Taesoo Kim.
NSDI 2017 Poster, Boston, MA, April 2017.
3. | **DistCoz: Tell Me What to Optimize in My Distributed Application**
Steffen Maass, **Mohan Kumar**, and Taesoo Kim.
NSDI 2017 Poster, Boston, MA, April 2017.
4. | **VNFStore: NFV State Externalizing Framework**
Mohan Kumar and Ada Gavrilovska.
Diversity Workshop at SOSP 2015, Monterey, CA, October 2015.

Awards and Recognition

EuroSys 2017	Best Student Paper Award	April 2017
SDN-NFV Security 2016	Best Paper Award	March 2016

Open Source Contributions

Zephyr Project	Contributed improvements to the Zephyr real-time operating system and protocol stack	2022
SLiRP	Bug fixes and enhancements to the SLiRP IPv6 networking module	2021

Travel Grants

1. | **25th ACM Symposium on Operating Systems Principles**
Monterey, CA
October 2015
2. | **Diversity Workshop at SOSP 2015**
Monterey, CA
October 2015
3. | **USENIX Symposium on Networked Systems Design and Implementation (NSDI)**
Boston, MA
March 2017

Invited Talks and Presentations

INFOCOM 2016	TCP Ordo: The Cost of Ordered Processing in TCP Servers	April 2016
ASPLOS 2018	LATR: Lazy Translation Coherence	March 2018

Work Experience

Research Scientist	Meta Reality Labs, Burlingame, CA Accelerating machine learning inference on embedded devices through optimization of CNN, RCNN, RNN, and transformer models for resource-constrained hardware platforms.	July 2019–Present
Software Engineering Intern	Meta, Menlo Park, CA Developed data center network system software using eBPF and XDP technologies.	May 2018–August 2018
Research Intern	AT&T Labs, Bedminster, NJ Improved per-core packet processing performance in software switching architectures.	June 2016–August 2016
Research Intern	HP Labs, Palo Alto, CA Developed communication software framework for next-generation data center systems.	May 2015–August 2015
Research Assistant	Georgia Institute of Technology, Atlanta, GA Conducted systems research under Dr. Ada Gavrilovska and Dr. Taesoo Kim in the CERCS and Systems Software & Security Lab.	May 2014–May 2019
Senior Software Engineer	Cisco Mobile Internet Technology Group, Bangalore, India Designed and developed software stack for 4G telecommunications nodes including Packet Data Network Gateway (PGW) and Serving Gateway (SGW) architectures.	August 2010–May 2014

Teaching Experience

Teaching Assistant	Georgia Institute of Technology, Atlanta, GA Teaching Assistant for Advanced Operating Systems (AOS) during Spring 2016.	January 2016–May 2016
Teaching Assistant	Georgia Institute of Technology, Atlanta, GA Teaching Assistant for Advanced Operating Systems (AOS) during Spring 2018.	January 2018–May 2018

Professional Service and Engagement

Program Committee	USENIX ATC 2020
External Reviewer	MLSys 2026

Additional Information

US Permanent Resident (Green Card holder, approved as outstanding researcher)